Tancrède AMIEL

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EDUCATION

CENTRALESUPÉLEC — Plateau de Saclay, FR
Diplôme d'ingénieur (Master's-level degree), General Engineering — French Grande École — *2024 – 2028*
GPA: 3.94/4.33 (Year 1), 4.03/4.33 (Year 2) — Gap year 2026–27 (industry placement in Tokyo)
Coursework: probability, statistics, convex optimisation, PDEs, machine learning, information theory, quantum computing, HPC

COLLÈGE STANISLAS — Paris, FR
Classes Préparatoires aux Grandes Écoles (CPGE) — PCSI, then PC (advanced stream)* — *2022 – 2024*
GPA: 3.92/4 (Year 1), 3.98/4 (Year 2) — intensive mathematics and physics programme for the Grandes Écoles exams

RESEARCH & ACADEMIC PROJECTS

TINYML IMITATION LEARNING – AutoGear — CentraleSupélec
Course: Entrepreneurship Track — Grade: 17/20 (A+) — *Jan. 2025 – Oct. 2026*
Universal automatic derailleur kit, built from scratch (cadence sensor, Arduino, servo), at 1/10 of Shimano's price — team of four

- **Obstacle:** no torque sensor (costliest part) → rider effort compensated statistically; TinyML on an Arduino
- **Method:** XGBoost (13.9 KB) predicts the gear 3 s ahead + hysteresis control, trained on 18.9 h of public rides
- **Result:** closed-loop replay: *within one gear of the rider 87% of the time* vs 81% for a torque-aware rule table

BAYESIAN OPTIMISATION & HPC – Blast-aware routing — CentraleSupélec – CEA-DAM
Course: Optimisation of Infrasonic Wave Detection for CTBT Verification — Grade: 19/20 (A+) — *Feb. – Apr. 2026*
Industrial research project with CEA-DAM (military applications division of France's atomic energy commission): route a device that may explode across a city at minimum expected damage — team of six

- **Method:** Bayesian optimisation: Gaussian process (Matérn 5/2) correcting a Kingery–Bulmash prior, active learning on route-relevant positions; parallel Δ-stepping (MPI), turn-aware A*
- **Result:** *19 of 10,000 runs* of ARMEN (2 h on 48 cores) in 2D, route uncertainty 14.5% → 4%; 3D route in < 10 s

QUANTUM-INSPIRED ML – Critique of Variational Tensor Neural Networks — CentraleSupélec
Course: Applications of Statistical and Quantum Physics to Information Science — *Jun. 2026*
Assessment of Jahromi & Orús (*Sci. Rep.*, 2024), MPO layers trained by DMRG sweeps: real novelty is *the weights' entanglement entropy readable during training*; accuracy and interpretability hold, scalability does not

PDE-CONSTRAINED OPTIMISATION – Acoustic absorber placement — CentraleSupélec
Course: Wave Control (applied functional analysis) — Grade: 15/20 (A) — *Nov. 2025*
Minimise a concert hall's acoustic energy with absorbing material → mineral wool on half of a level-2 pre-fractal wall — team of six; designed the source equations, proved well-posedness (Fredholm alternative), derived the adjoint gradient

COMBINATORICS RESEARCH – French Tournament of Young Mathematicians — Animath – TFJM^2
National team research competition for secondary-school students — 3rd place, regional round — *Jan. – May 2022*
Five months on an open problem: algorithms turned into formal proofs, report reviewed by researchers, defended before a jury

PROFESSIONAL EXPERIENCE

AI / ML ENGINEER (INTERN) – AI Solution Design & Deployment — Tokyo, JAPAN
Dassault Systèmes K.K. — AI team — *Sep. 2026 – Feb. 2027*
Internal forward-deployed AI engineer driving generative-AI adoption, from use-case prioritisation to open-weight deployment

- **Process automation (RFx):** OKF and vector-DB restructuring, RAG/OKF-RAG, web search, JA–EN translation, MCP
- **Physical AI Lab:** launching the lab, strategic and hands-on: stack blueprint, hardware costing, curriculum, launch projects
- **On-premise open-weight models:** GPU server choice and costing from executive needs and budget

DEPUTY PROJECT MANAGER & SOLUTION ARCHITECT – IAgriM — Paris – Bamako, MALI
Danew Talla Electronics — *Jul. – Aug. 2024*
Two-mode AI assistant for Malian farmers (offline knowledge-base routing; online speech-to-text and LLM), overcoming poor connectivity, spoken-only languages and low digital literacy; *approved by the Ministers of Digital Economy and of Employment of Mali*; consortium with CIAR and the NGO AgriCap; on hold pending funding

TEACHING

MATHS ORAL EXAMINER (Collège Stanislas) & PRIVATE TUTOR — Paris, FR
About 50 students examined at the board, 2–4 h/week; 10+ tutees in mathematics and physics — *Sep. 2024 – Jun. 2026*

SKILLS

Languages	French (native) — English (C1/C2, bilingual) — Spanish (B2, DELE certified)
Programming	Python (Pandas, scikit-learn, XGBoost, TensorFlow), MPI, SLURM, Git, Docker, Kubernetes, C++